
Forecasting

FORECASTING

- ❑ A statement about the future value of a variable of interest.
Example: demand forecast.
- ❑ Forecasts are made with reference to a specific time horizon.
- ❑ Underlying basis of all business decisions
 - Production
 - Inventory
 - Personnel
 - Facilities



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Types of Forecasts by Time Horizon

- ❑ Short-range forecast
 - Up to 1 year; usually less than 3 months
 - Job scheduling, worker assignments
- ❑ Medium-range forecast
 - 3 months to 3 years
 - Sales & production planning, budgeting
- ❑ Long-range forecast
 - 3+ years
 - New product planning, facility location

Uses of Forecasts

Forecasts affect decisions and activities throughout an organization

Accounting	Cost/profit estimates
Finance	Cash flow and funding
Human Resources	Hiring/recruiting/training
Marketing	Pricing, promotion, strategy
MIS	IT/IS systems, services
Operations	Schedules, MRP, workloads
Product/service design	New products and services

Elements of a Good Forecast

A properly prepared forecast should fulfill certain requirements:

1. The forecast should be **timely. Usually, a certain amount of time is needed to respond to** the information contained in a forecast. For example, capacity cannot be expanded overnight, nor can inventory levels be changed immediately. Hence, the forecasting horizon must cover the time necessary to implement possible changes.
 2. The forecast should be **accurate, and the degree of accuracy should be stated. This will** enable users to plan for possible errors and will provide a basis for comparing alternative forecasts.
 3. The forecast should be **reliable; it should work consistently. A technique that sometimes** provides a good forecast and sometimes a poor one will leave users with the uneasy feeling that they may get burned every time a new forecast is issued.
 4. The forecast should be expressed in **meaningful units. Financial planners need to know** how many dollars will be needed, production planners need to know how many units will be needed, and schedulers need to know what machines and skills will be required. The choice of units depends on user needs.
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Elements of a Good Forecast

5. The forecast should be **in writing**. **Although this will not guarantee that all concerned** are using the same information, it will at least increase the likelihood of it. In addition, a written forecast will permit an objective basis for evaluating the forecast once actual results are in.

6. The forecasting technique should be **simple to understand and use**. **Users often lack** confidence in forecasts based on sophisticated techniques; they do not understand either the circumstances in which the techniques are appropriate or the limitations of the techniques. Misuse of techniques is an obvious consequence. Not surprisingly, fairly simple forecasting techniques enjoy widespread popularity because users are more comfortable working with them.

7. The forecast should be **cost-effective: The benefits should outweigh the costs**. that they may get burned every time a new forecast is issued.

Steps in the Forecasting Process

- 1. Determine the purpose of the forecast.** How will it be used and when will it be needed? This step will provide an indication of the level of detail required in the forecast, the amount of resources (personnel, computer time, dollars) that can be justified, and the level of accuracy necessary.
- 2. Establish a time horizon.** The forecast must indicate a time interval, keeping in mind that accuracy decreases as the time horizon increases.
- 3. Obtain, clean, and analyze appropriate data.** Obtaining the data can involve significant effort. Once obtained, the data may need to be “cleaned” to get rid of outliers and obviously incorrect data before analysis.
- 4. Select a forecasting technique.**
- 5. Make the forecast.**
- 6. Monitor the forecast.** A forecast has to be monitored to determine whether it is performing in a satisfactory manner. If it is not, re-examine the method, assumptions, validity of data, and so on; modify as needed; and prepare a revised forecast.

Types of Forecasts

- ❑ **Qualitative/Judgmental forecasts rely on analysis of subjective inputs obtained from various sources**, such as consumer surveys, the sales staff, managers and executives, and panels of experts. Quite frequently, these sources provide insights that are not otherwise available.
 - ❑ **Quantitative/Time-series forecasts simply attempt to project past experience into the future. These** techniques use historical data with the assumption that the future will be like the past. Some models merely attempt to smooth out random variations in historical data; others attempt to identify specific patterns in the data and project or extrapolate those patterns into the future, without trying to identify causes of the patterns.
 - ❑ **Associative models use equations that consist of one or more explanatory variables that** can be used to predict demand. For example, demand for paint might be related to variables such as the price per gallon and the amount spent on advertising, as well as to specific characteristics of the paint (e.g., drying time, ease of cleanup).
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Judgmental Forecasts

Executive opinions: A small group of upper-level managers (e.g., in marketing, operations, and finance) may meet and collectively develop a forecast. This approach is often used as a part of long-range planning and new product development. It has the advantage of bringing together the considerable knowledge and talents of various managers. However, there is the risk that the view of one person will prevail, and the possibility that diffusing responsibility for the forecast over the entire group may result in less pressure to produce a good forecast.

Judgmental Forecasts

Sales force opinions: Members of the sales staff or the customer service staff are often good sources of information because of their **direct contact with consumers**. They are often aware of any plans the customers may be considering for the future. There are, however, several drawbacks to using sales force opinions. One is that staff members may be unable to distinguish between what customers **would like to do and what they actually will do**.

Another is that these people are sometimes overly influenced by **recent experiences**. Thus, after several periods of low sales, their estimates may tend to **become pessimistic**. After several periods of good sales, they may tend to **be too optimistic**.

In addition, if forecasts are used to establish sales quotas, there will be a conflict of interest because it is to the salesperson's advantage to provide low sales estimates.

Judgmental Forecasts

Consumer surveys: Because it is the consumers who ultimately determine demand, it seems natural to solicit input from them. In some instances, every customer or potential customer can be contacted. However, usually there are too many customers or there is no way to identify all potential customers. Therefore, organizations seeking consumer input usually resort to consumer surveys, which enable them to sample consumer opinions. The obvious advantage of consumer surveys is that they can tap information that might not be available elsewhere. On the other hand, a considerable amount of knowledge and skill is required to construct a survey, administer it, and correctly interpret the results for valid information. Surveys can be expensive and time-consuming. In addition, even under the best conditions, surveys of the general public must contend with the possibility of irrational behavior patterns. For example, much of the consumer's thoughtful information gathering before purchasing a new car is often undermined by the glitter of a new car showroom or a high-pressure sales pitch. Along the same lines, low response rates to a mail survey should—but often don't—make the results suspect. If these and similar pitfalls can be avoided, surveys can produce useful information.

Judgmental Forecasts

- ❖ Another approach is the **Delphi method**, an **iterative process** intended to **achieve a consensus** forecast.
- ❖ This method involves **circulating a series of questionnaires** among individuals who possess the knowledge and ability to contribute meaningfully.
- ❖ Responses are kept **anonymous**, which tends to **encourage honest responses and reduces the risk** that one person's opinion will prevail.
- ❖ Each **new** questionnaire is developed using the information extracted from the **previous** one, thus enlarging the scope of information on which participants can base their judgments.
- ❖ The Delphi method has been applied to a variety of situations, not all of which involve forecasting. The discussion here is limited to its use as a forecasting tool.



Judgmental Forecasts

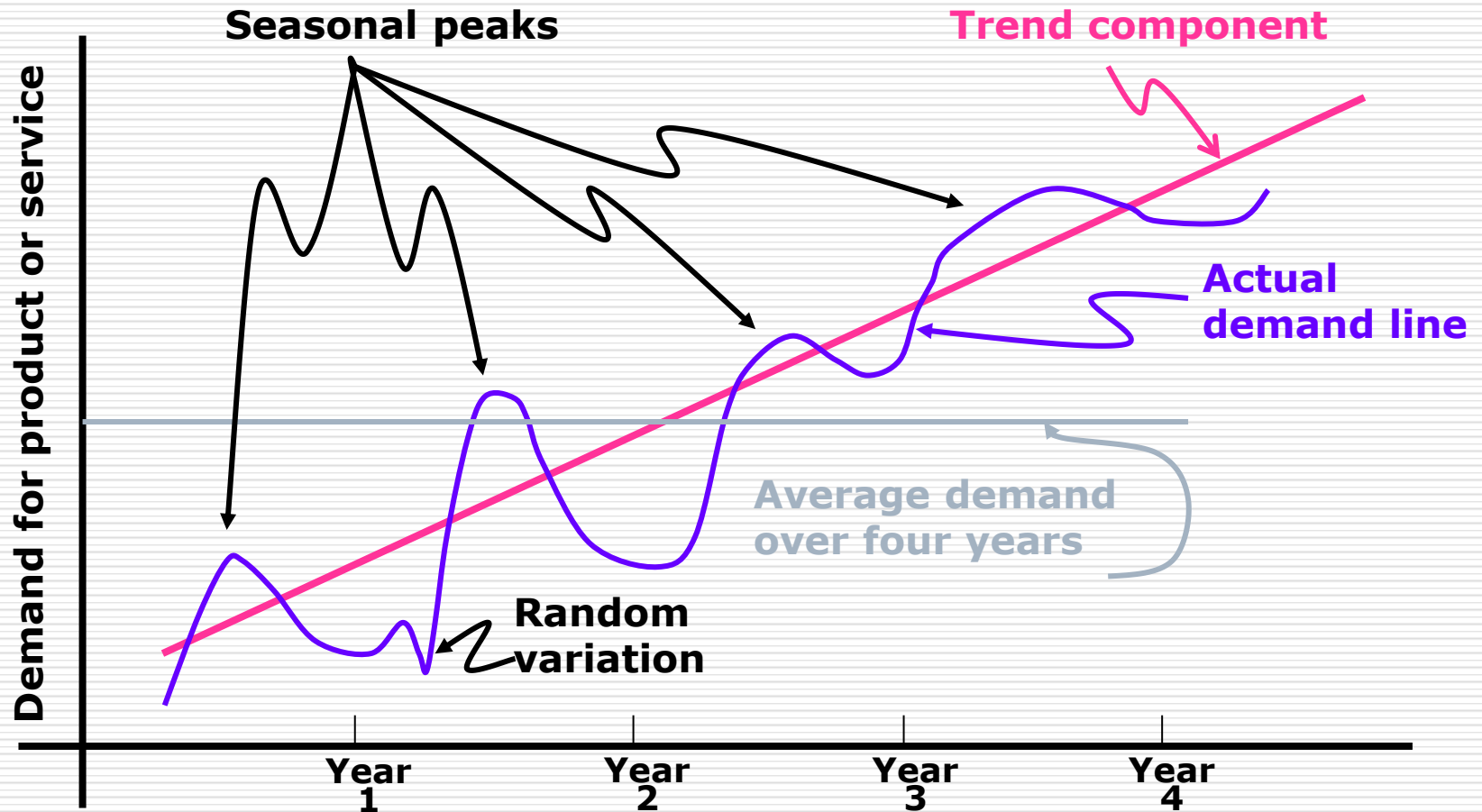
- ❖ As a forecasting tool, the Delphi method is useful for technological forecasting, that is, for **assessing changes in technology** and their **impact on an organization**.
 - ❖ Often the **goal is to predict** when a certain event will occur.
 - ❖ For instance, the goal of a Delphi forecast might be to predict when video telephones might be installed in at least 50 percent of residential homes or when a vaccine for a disease might be developed and ready for mass distribution.
 - ❖ For the most part, these are long-term, single-time forecasts, which usually have very little hard information to go by or data that are costly to obtain, so the problem does not lend itself to analytical techniques. Rather, judgments of experts or others who possess sufficient knowledge to make predictions are used.
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Time series Forecasts

Time series: A time-ordered sequence of observations taken at regular intervals. The behaviours of the time series data can be described as follows:

- Trend - A long-term upward or downward movement in data.
 - Seasonality - Short-term regular variations in data related to the calendar or time of day.
 - Cycle - Wavelike variations lasting more than one year.
 - Irregular variations - Caused by unusual circumstances, not reflective of typical behavior.
 - Random variations - caused by chance, Residual variations after all other behaviors are accounted for.
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Product Demand Charted over 4 Years with Trend and Seasonality



Naive Forecasts

The forecast for any period equals the previous period's actual value.

Characteristics of naive forecast:

- Simple to use
 - Virtually no cost
 - Quick and easy to prepare
 - Data analysis is nonexistent
 - Easily understandable
 - Cannot provide high accuracy
 - Can be a standard for accuracy
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Techniques for Averaging

- ☐ Moving average
 - ☐ Weighted moving average
 - ☐ Exponential smoothing
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Moving Averages

- **Moving average** – A technique that averages a number of recent actual values, updated as new values become available.

- For ***n*** period moving average
 $n = 2, 3, 4, 5, \dots$

$$F_t = \frac{\sum_{i=1}^n A_i}{n}$$

- **Weighted moving average** – More recent values in a series are given more weight in computing the forecast.

- w is the respective weight

$$F_t = \frac{\sum_{i=1}^n A_i w_i}{\sum_{i=1}^n w_i}$$

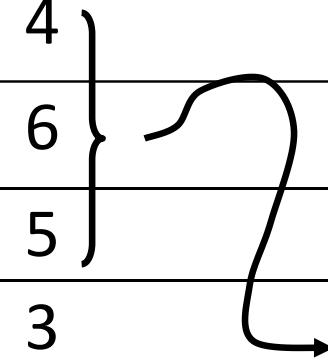
Moving Average Example

You're manager of a museum store that sells historical replicas. You want to forecast sales (000) for 2003 using a 3-period moving average.

1998	4
1999	6
2000	5
2001	3
2002	7

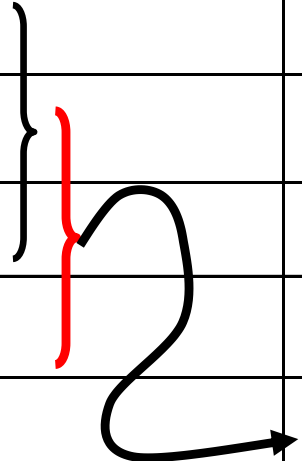
Moving Average Solution

Time	Response Y_i	Moving Total ($n=3$)	Moving Average ($n=3$)
1998	4	NA	NA
1999	6	NA	NA
2000	5	NA	NA
2001	3	4+6+5=15	15/3 = 5
2002	7		
2003	NA		



Moving Average Solution

Time	Response Y_i	Moving Total ($n=3$)	Moving Average ($n=3$)
1998	4	NA	NA
1999	6	NA	NA
2000	5	NA	NA
2001	3	$4+6+5=15$	$15/3 = 5$
2002	7	$6+5+3=14$	$14/3=4 \frac{2}{3}$
2003	NA		



Moving Average Solution

Time	Response Y_i	Moving Total ($n=3$)	Moving Average ($n=3$)
1998	4	NA	NA
1999	6	NA	NA
2000	5	NA	NA
2001	3	4+6+5=15	15/3=5.0
2002	7	6+5+3=14	14/3=4.7
2003	NA	5+3+7=15	15/3=5.0

Example of moving average and weighted moving average

Example:

<u>Period</u>	<u>Demand</u>	<u>Forecast</u>
1	42	
2	40	42
3	43	
4	40	
5	42	

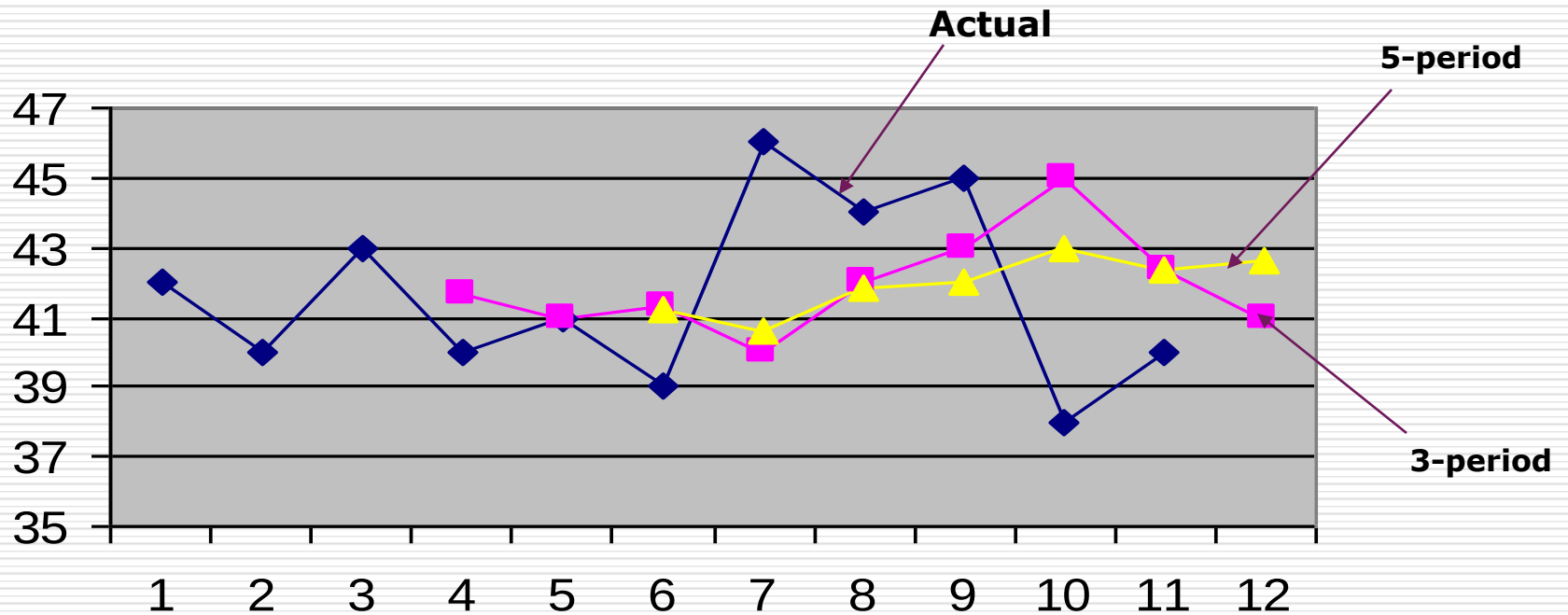
Simple moving average (4 months)

$$F_6 = (40 + 43 + 40 + 42) / 4 = 41.25$$

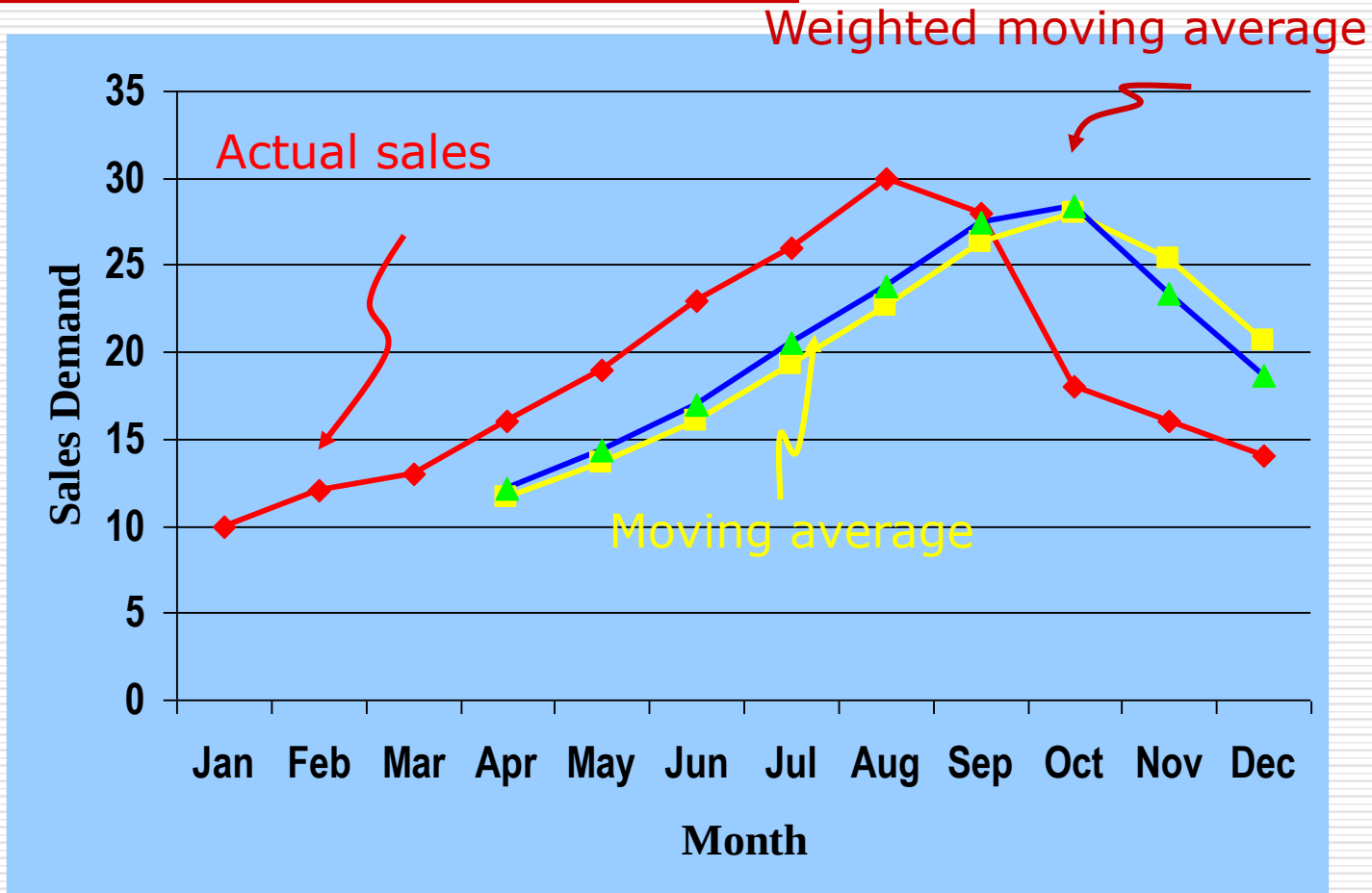
- Compute a weighted average forecast using a weight of 0.40 for the most recent period, 0.30 for the next most recent, 0.20 for the next, and 0.10 for the next.
- Weighted moving average (4 months)

$$F_6 = \frac{(42 * 0.4 + 40 * 0.3 + 43 * 0.2 + 40 * 0.1)}{(0.4 + 0.3 + 0.2 + 0.1)} = 41.4$$

Simple Moving Average



Actual Demand, Moving Average, Weighted Moving Average



Exponential Smoothing

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

Next forecast = Previous forecast + α $\underbrace{(\text{Actual demand} - \text{Previous forecast})}_{\text{Forecasting error}}$

α $\longrightarrow$ Percentage of error / smoothing constant

A-F is the error term, α is the % feedback

Exponential Smoothing

Exponential smoothing ($\alpha = 0.1$)

$$F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$$

$$\begin{aligned} F_3 &= F_2 + \alpha (A_2 - F_2) \\ &= 42 + 0.1 (40 - 42) = 41.8 \end{aligned}$$

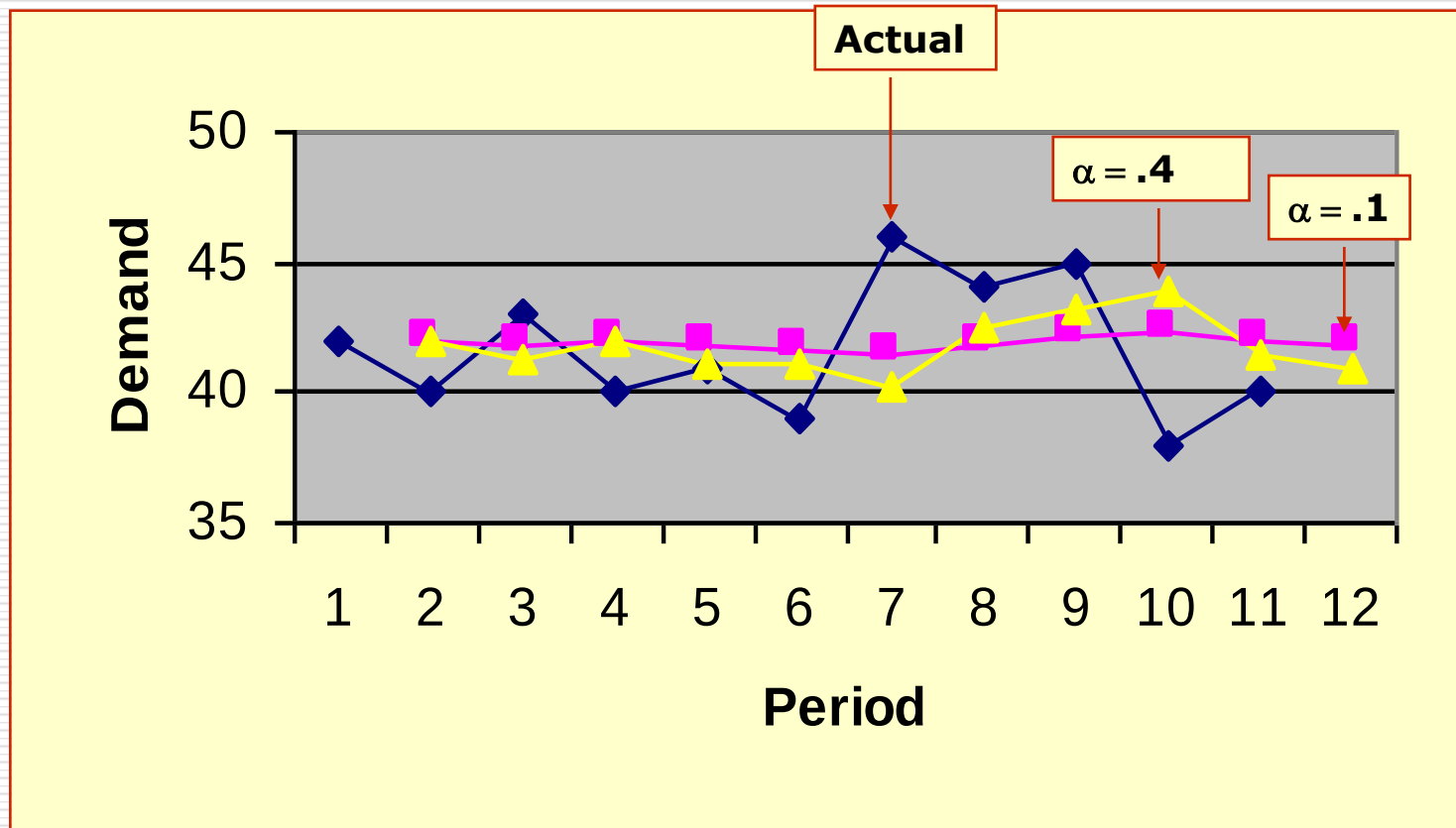
$$\begin{aligned} F_4 &= F_3 + \alpha (A_3 - F_3) \\ &= 41.8 + 0.1 (43 - 41.8) = 41.92 \end{aligned}$$

$$\begin{aligned} F_5 &= F_4 + \alpha (A_4 - F_4) \\ &= 41.92 + 0.1 (40 - 41.92) = 41.728 \end{aligned}$$

Example 3 - Exponential Smoothing

Actual	Alpha = 0.1	Error	Abs	Alpha = 0.4	Error	Abs
42						
40	42	-2.00	2.00	42	-2	2
43	41.8	1.20	1.20	41.2	1.8	1.8
40	41.92	-1.92	1.92	41.92	-1.92	1.92
41	41.73	-0.73	0.73	41.15	-0.15	0.15
39	41.66	-2.66	2.66	41.09	-2.09	2.09
46	41.39	4.61	4.61	40.25	5.75	5.75
44	41.85	2.15	2.15	42.55	1.45	1.45
45	42.07	2.93	2.93	43.13	1.87	1.87
38	42.36	-4.36	4.36	43.88	-5.88	5.88
40	41.92	-1.92	1.92	41.53	-1.53	1.53
	41.73		4.45	40.92		2.44

Picking a Smoothing Constant



Example:

Forecasts based on averages. Given the following data:

Period	Number of Complaints
1	60
2	65
3	55
4	58
5	64

Prepare a forecast for period 6 using each of these approaches:

- The appropriate naive approach.
 - A three-period moving average.
 - A weighted average using weights of .50 (most recent), .30, and .20.
 - Exponential smoothing with a smoothing constant of .40.
-

Example:

- Plot the data to see if there is a pattern. Variations around an average (i.e., no trend or cycles). Therefore, the most recent value of the series becomes the next forecast: 64.
- Use the latest values. $MA_3 = \frac{55 + 58 + 64}{3} = 59$
- $F = .20(55) + .30(58) + .50(64) = 60.4$
- Start with period 2. Use the data in period 1 as the forecast for period 2, and then use exponential smoothing for successive forecasts.

Period	Number of Complaints	Forecast	Calculations
1	60		
2	65	60	[The previous value of the series is used as the starting forecast.]
3	55	62	$60 + .40(65 - 60) = 62$
4	58	59.2	$62 + .40(55 - 62) = 59.2$
5	64	58.72	$59.2 + .40(58 - 59.2) = 58.72$
6		60.83	$58.72 + .40(64 - 58.72) = 60.83$

Associative Forecasting

Predictor variables - used to predict values of variable interest

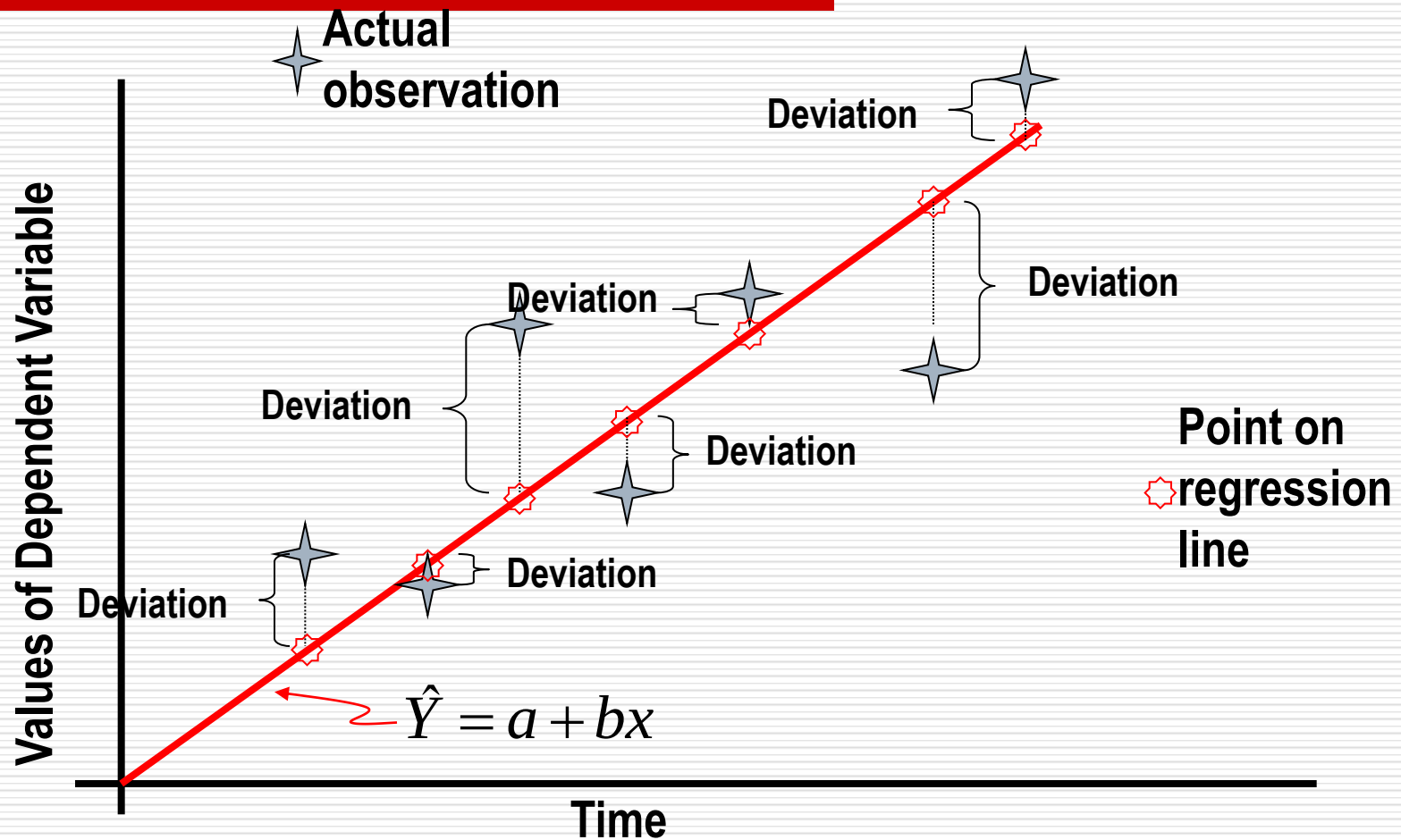
Regression - technique for fitting a line to a set of points

Least squares line - The least squares regression method is a **mathematical technique** that finds the **best-fitting line or curve** for **a set of data points**, providing a visual demonstration of the relationship between the data points. Each point of data represents the relationship between a known independent variable and an unknown dependent variable. This method is commonly used by statisticians and traders who want to identify trading opportunities and trends. Least squares regression is used to predict the behavior of dependent variables.

The method works by minimizing the sum of the offsets or residuals of points from the plotted curve.

Linear Regression

Least Squares Method

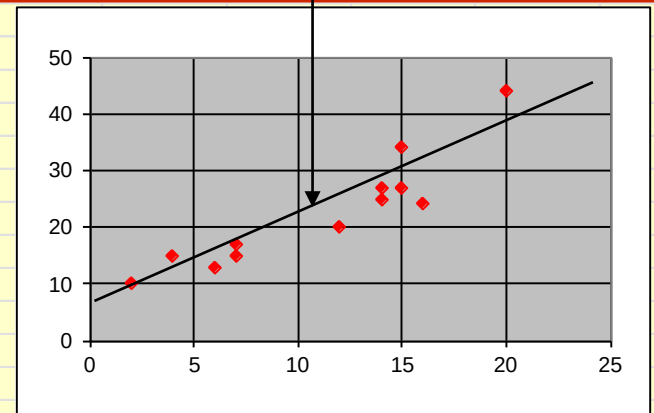


Linear Regression

No. of obs	Unit Sales, x (in \$ millions)	Profits, y (in \$ millions)
1	\$ 7	\$0.15
2	2	0.10
3	6	0.13
4	4	0.15
5	14	0.25
6	15	0.27
7	16	0.24
8	12	0.20
9	14	0.27
10	20	0.44
11	15	0.34
12	7	0.17

A straight line is fitted to a set of sample points.

Computed relationship



Linear Regression

$$y = a + bx$$

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$a = \bar{y} - b\bar{x}$$

n= number of observations

Linear Regression

No. of obs	Unit Sales, x (in \$ millions)	Profits, y (in \$ millions)	xy	x ²
1	\$ 7	\$0.15	1.05	49
2	2	0.10	0.2	4
3	6	0.13	0.78	36
4	4	0.15	0.6	16
5	14	0.25	3.5	196
6	15	0.27	4.05	225
7	16	0.24	3.84	256
8	12	0.20	2.4	144
9	14	0.27	3.78	196
10	20	0.44	8.8	400
11	15	0.34	5.1	225
12	7	0.17	1.19	49
-	$\sum x = 132$	$\sum y = 2.71$	$\sum xy = 35.29$	$\sum x^2 = 1796$

Linear Regression

$$b = \frac{12 * 35.29 - 132 * 2.71}{12 * 1796 - (132)^2}$$

$$\Rightarrow b = 0.0159$$

Linear Regression

$$b = \frac{12 * 35.29 - 132 * 2.71}{12 * 1796 - (132)^2}$$

$$\Rightarrow b = 0.0159$$

$$a = \bar{y} - b\bar{x}$$

$$\Rightarrow a = \frac{2.71}{12} - 0.0159 \frac{132}{12}$$

$$\Rightarrow a = 0.0506$$

Linear Regression

$$b = \frac{12 * 35.29 - 132 * 2.71}{12 * 1796 - (132)^2}$$

$$\Rightarrow b = 0.0159$$

$$a = \bar{y} - b\bar{x}$$

$$\Rightarrow a = \frac{2.71}{12} - 0.0159 \frac{132}{12}$$

$$\Rightarrow a = 0.0506$$

$$y = 0.0506 + 0.0159 x$$

Linear Regression Assumptions

- ❑ Variations around the line are random
 - ❑ Deviations around the line normally distributed
 - ❑ Predictions are being made only within the range of observed values
 - ❑ For best results:
 - Always plot the data to verify linearity
 - Check for data being time-dependent
 - Small correlation may imply that other variables are important
-

Forecast Accuracy

- ❑ Error - difference between actual value and predicted value
 - ❑ Mean Absolute Deviation (MAD)
 - Average absolute error
 - Easy to compute
 - Weights errors linearly
 - ❑ Mean Squared Error (MSE)
 - Average of squared error
 - Squares error
 - More weight to large errors
 - ❑ Mean Absolute Percent Error (MAPE)
 - Average absolute percent error
-

MAD, MSE and MAPE

$$\text{MAD} = \frac{\sum |\text{Actual} - \text{forecast}|}{n}$$

$$\text{MSE} = \frac{\sum (\text{Actual} - \text{forecast})^2}{n - 1}$$

$$\text{MAPE} = \frac{\sum (|\text{Actual} - \text{forecast}| / \text{Actual}) * 100}{n}$$

Example

[illegible]

Example

[illegible]

Example

Period	Actual	Forecast	(A-F)	A-F	(A-F)^2	(A-F /Actual)*100
1	217	215	2	2		
2	213	216	-3	3		
3	216	215	1	1		
4	210	214	-4	4		
5	213	211	2	2		
6	219	214	5	5		
7	216	217	-1	1		
8	212	216	-4	4		

Example

Period	Actual	Forecast	(A-F)	A-F	(A-F)^2	(A-F /Actual)*100
1	217	215	2	2	4	
2	213	216	-3	3	9	
3	216	215	1	1	1	
4	210	214	-4	4	16	
5	213	211	2	2	4	
6	219	214	5	5	25	
7	216	217	-1	1	1	
8	212	216	-4	4	16	

Example

Period	Actual	Forecast	(A-F)	A-F	(A-F)^2	(A-F /Actual)*100
1	217	215	2	2	4	0.92
2	213	216	-3	3	9	1.41
3	216	215	1	1	1	0.46
4	210	214	-4	4	16	1.90
5	213	211	2	2	4	0.94
6	219	214	5	5	25	2.28
7	216	217	-1	1	1	0.46
8	212	216	-4	4	16	1.89

Example

Period	Actual	Forecast	(A-F)	A-F	(A-F)^2	(A-F /Actual)*100
1	217	215	2	2	4	0.92
2	213	216	-3	3	9	1.41
3	216	215	1	1	1	0.46
4	210	214	-4	4	16	1.90
5	213	211	2	2	4	0.94
6	219	214	5	5	25	2.28
7	216	217	-1	1	1	0.46
8	212	216	-4	4	16	1.89
			-2	22	76	10.26
MAD=	2.75					
MSE=	10.86					
MAPE=	1.28 %					